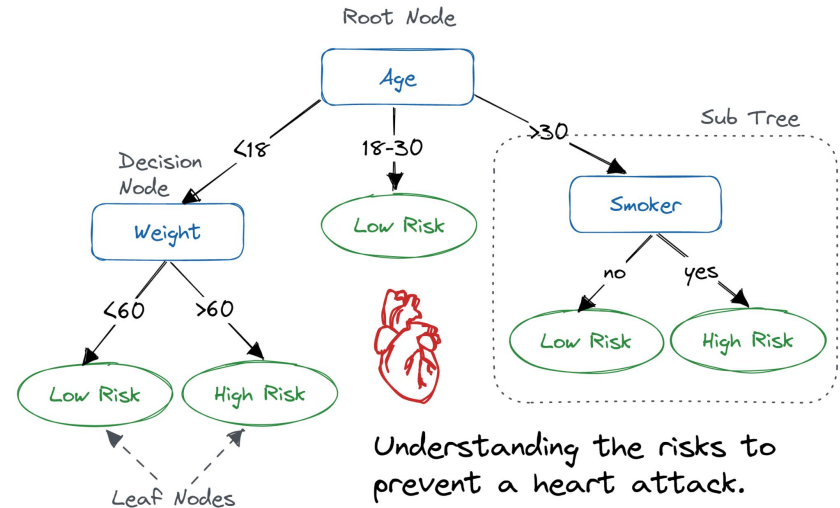

Decision Trees, Random Forests & Support Vector Machines

By: Sebastián Avilez Hernández, Cristian Rolando
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Decision Trees

Decision Trees

Decision trees represent supervised learning algorithms employed for both classification and regression tasks and resemble flowcharts in structure. They begin with a root node containing a specific data query, which leads to branches holding possible responses, these branches connect to internal decision nodes that pose additional questions, generating further outcomes. Machine learning decision trees can function as either classification or regression trees, often abbreviated as CART.



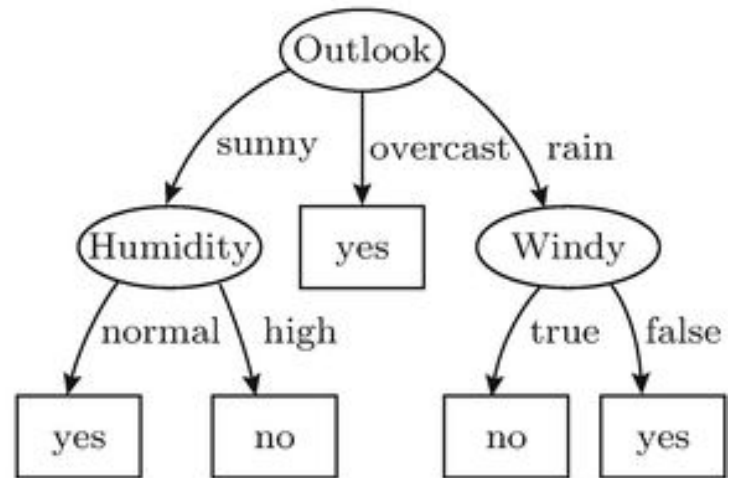
Data

Decision trees excel at processing diverse data types without extensive preparation, unlike alternative machine learning approaches requiring data scaling or transformation. Three primary strategies exist for managing missing data in decision tree models: removal, substitution, and adaptive learning.

Student	Prior Experience	Course	Time	Liked
1	Yes	Programming	Day	Yes
2	No	Programming	Day	No
3	Yes	History	Night	No
4	No	Programming	Night	Yes
5	Yes	English	Day	Yes
6	No	Programming	Day	No
7	Yes	Programming	Day	No
8	Yes	Mathematics	Night	Yes
9	Yes	Programming	Night	Yes
10	Yes	Programming	Night	No

How it work

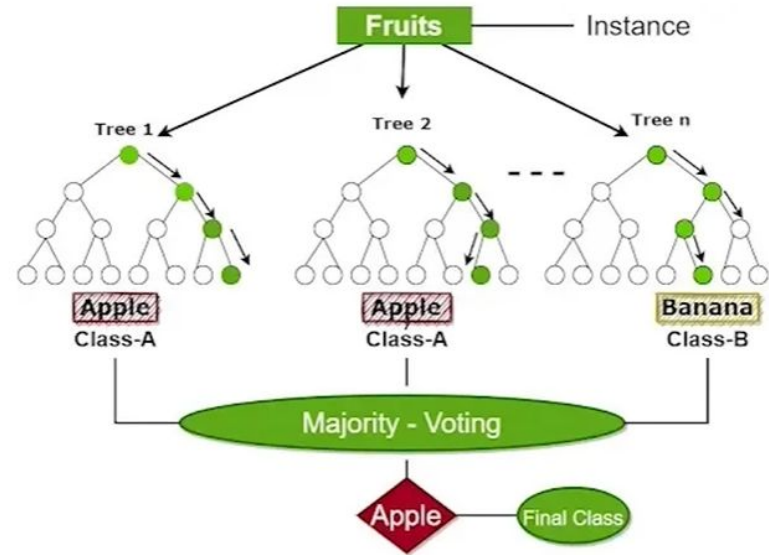
<i>Outlook</i>	<i>Temp</i>	<i>Humidity</i>	<i>Windy</i>	<i>Golf?</i>
rainy	hot	high	false	no
rainy	hot	high	true	no
overcast	hot	high	false	yes
sunny	mild	high	false	yes
sunny	cool	normal	false	yes
sunny	cool	normal	true	no
overcast	cool	normal	true	yes
rainy	mild	high	false	no
rainy	cool	normal	false	yes
sunny	mild	normal	false	yes
rainy	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
sunny	mild	high	true	no



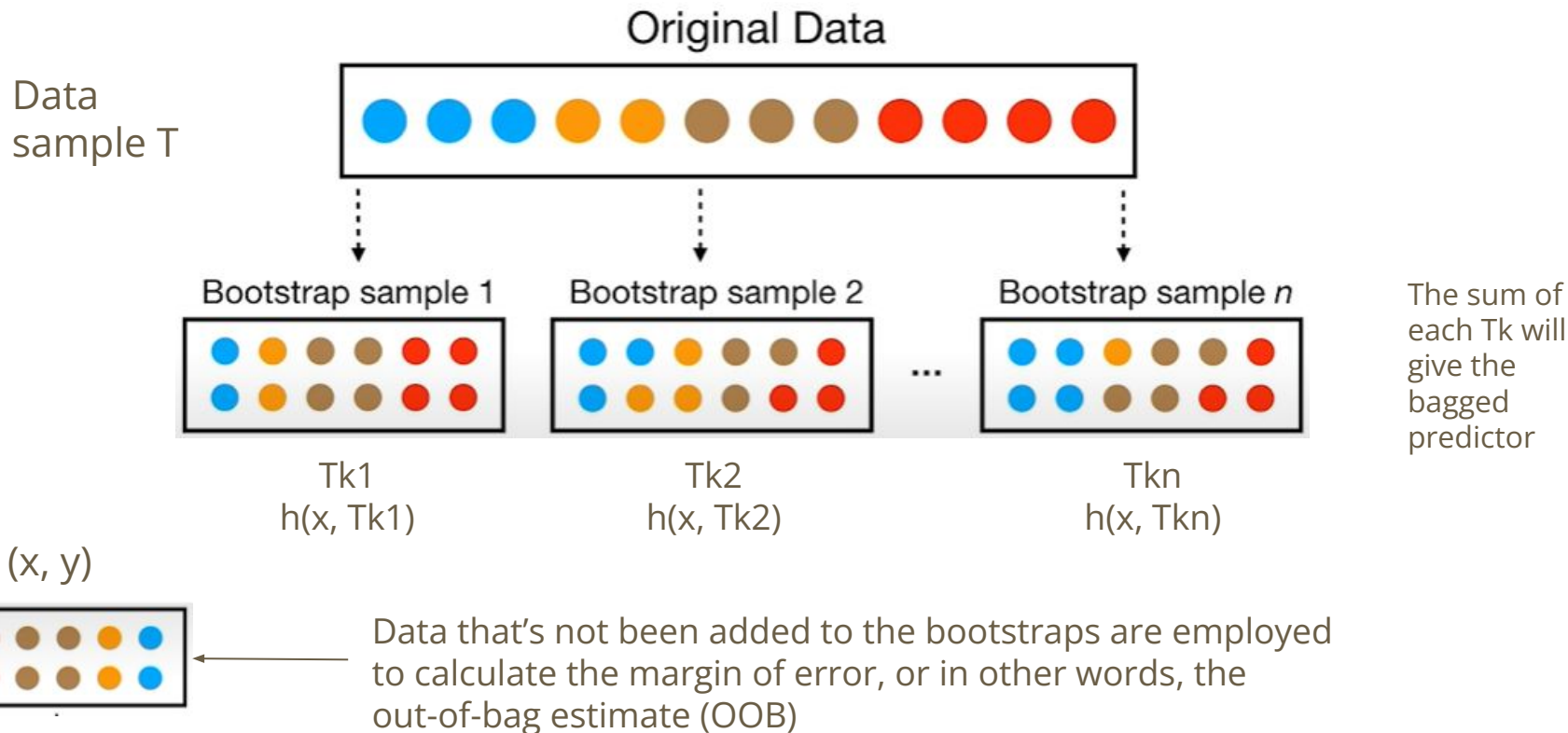
Random Forest Algorithm

Random Forests

- Instead of one tree, now there are several
- Reduction of over-fitness due to covering bigger datasets
- Purposeful for both classification and regression data
- Great for calculating both accurate results and estimate averages with little to no result bias.
- **Category of ML: supervised learning**



How does it work?



Industry applications



Retail & E - Commerce

Great for product categorization
and personalized marketing

Finance

Fraud detection and checking credit
history

Healthcare

Detection of diseases based on
symptoms and drug response
modeling

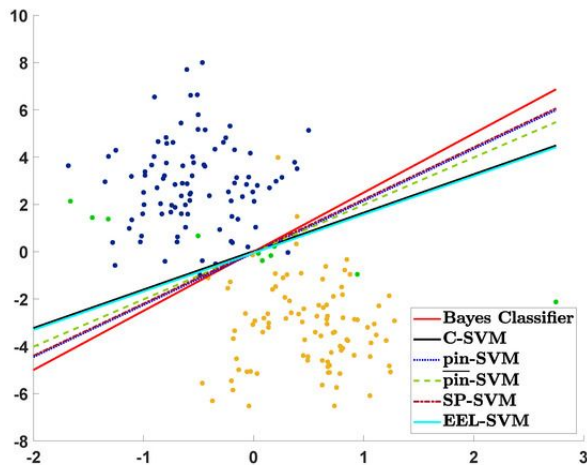
Quality control

Supply chain optimization for faulty
products



Support Vector Machine (SVM)

SVMs offer robust classification capabilities for diverse datasets.



(a)

Key Strengths of SVMs



Maximizing Margin

Optimal decision boundary



Kernel Trick

Handles non-linear data



Regularization

Manages overfitting

SVMs achieve high accuracy and generalization through margin maximization, kernel functions for non-linear data, and robust regularization.



Real-World Impact

Medical Diagnosis

Precise cancer detection

Spam Detection

Email classification

Financial Fraud

Risk assessment

SVMs are critical in fields demanding high precision, from medical diagnosis to cybersecurity.